

Analysis Plan

Predicting Billboard 200 Success from Album Cover Art

Music Capstone Group

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1 Introduction and Research Question

Due to the advent of new internet technologies, online streaming has become the most popular way for people to listen to the music that they want. Companies such as Spotify are worth billions of dollars and are projected to continue to grow. Alongside them are music artists and their producers and record labels who still aim to create the most popular songs. While much analysis has been done on the audio characteristics of hit songs, an area which remains under researched is how album cover art impacts the performance of songs. On music streaming platforms, album cover art acts as a visual placeholder for a song and is prominently featured when a song is playing. Cover art can act as a user's first impression of an album and influence users to listen by capturing their attention for example. As such, this study investigates whether one can predict whether an album will remain in the Billboard 200 based on visual characteristics of its cover art.

2 Challenges and Alternative Proposals

This study considered several other ways to measure the importance of cover art but decided that they were not worth pursuing. They are outlined in this section.

2.1 Measuring Song Success

Initially, this study was aimed at using album cover art to predict the performance of a song in the Billboard Hot 100. However, this ran into a problem: all the songs in an album share the same cover art but can have drastically different performances. If multiple songs from the same album made the Hot 100 this would result in problems for our model which wouldn't

be able to use the album cover art to explain the differences in performance. Additionally, if this study were to try and predict whether a song made the Hot 100 or not, sampling multiple songs from the same album that made or didn't make the Hot 100 would result in the same problem.

2.2 Predicting Based on Song Audio Features

Initially, this study considered predicting song performance based on audio features collected from the Spotify API. Unfortunately, Spotify deprecated their audio features API endpoint, making it impossible to get audio features that way. This study investigated other sources of audio features, but none were as simple, comprehensive, or free like Spotify's. For these reasons, this study moved to investigating cover art.

2.3 Predicting Based on Album Audio Features

Another intuitive idea is to incorporate audio features into our analysis and find patterns with visual features. However, finding album wide audio features poses many challenges. First is that albums often don't have consistent audio features. Individual songs in an album can vary dramatically, and averaging measurements like beats per minute can lose information. Second is that given the difficulty in defining album audio features, there are very few data sources that provide album level audio data, making it impractical to pursue. We searched for album level genre data and found getgenre.com, however, approximately 25% of albums from the Billboard 200 do not have a genre associated with them, so we abandoned this approach.

3 Data Collection

3.1 Data Sources

We combine data from two primary sources:

(a) Album Cover Images and Metadata (MusicBrainz API)

We use the MusicBrainz API to access a comprehensive database of album releases. Critically, this database includes *all albums*—not just those that appeared on the Billboard 200. This allows us to construct a dataset with:

- **Positive cases (hits):** Albums that appeared on the Billboard 200

- **Negative cases (non-hits):** Albums that were released but never appeared on the Billboard 200

For each album, we obtain:

- Album cover image (RGB format)
- Album metadata (title, artist, release date, album ID)

(b) **Billboard 200 Chart Data**

Weekly Billboard 200 chart positions used to identify which albums from our MusicBrainz dataset achieved chart success. We match albums between MusicBrainz and Billboard using album title, artist name, and release date.

Key advantage of this approach: By using MusicBrainz as our primary source, we avoid selection bias that would occur if we only analyzed albums that charted. Our dataset includes the full population of released albums, allowing us to build a true binary classifier (hit vs. non-hit) rather than only comparing relative success among hits.

3.2 Dataset Structure

Each observation in our dataset represents a single album and contains:

- Album cover image (resized to 180×180 pixels for consistency)
- Album metadata (title, artist, release date, MusicBrainz ID)
- Response variable: Binary indicator of Billboard 200 appearance

$$Y_i = \begin{cases} 1 & \text{if album } i \text{ appeared on Billboard 200} \\ 0 & \text{otherwise} \end{cases}$$

3.3 Sample Size and Prototyping

Current prototype: We are developing our methodology using a sample of 315 albums drawn from MusicBrainz. This sample includes:

- Albums that appeared on the Billboard 200 (positive cases)
- Albums that did not appear on the Billboard 200 (negative cases)

This sample size allows for efficient prototyping of feature extraction, clustering, and model development.

Full analysis: Once methodology is finalized, we will scale to the complete matched dataset of all MusicBrainz albums with available chart information. The exact size will depend on:

- Availability of high-quality cover art images
- Successful matching between MusicBrainz and Billboard databases
- Time period selected for analysis (e.g., 2018–2024)

4 Feature Engineering

Visual features are extracted from each album cover image and organized into four feature groups. This organization is critical for our analysis approach, as different feature types require different distance metrics and preprocessing strategies.

4.1 Feature Groups

We extract 70 total features organized into four families:

4.1.1 RGB Features (12 features)

Statistical moments for each color channel:

- Mean, standard deviation, skewness, and kurtosis for Red, Green, and Blue channels
- Captures overall color intensity and distribution characteristics

4.1.2 HSV Features (24 features)

Color histograms in HSV (Hue-Saturation-Value) space:

- 8-bin histograms for each of Hue, Saturation, and Value
- Represents color distribution in a perceptually-meaningful space
- Data type: Histogram/compositional data (proportions)

4.1.3 Texture Features (32 features)

Local Binary Pattern (LBP) histogram:

- 32-bin histogram of texture patterns
- Captures micro-texture characteristics (smooth vs. rough, uniform vs. varied)
- Data type: Frequency counts

4.1.4 Edge Features (2 features)

Edge detection measures:

- `sobel_mean`: Average edge gradient magnitude across the image
- `edge_density`: Proportion of pixels classified as edges
- Captures visual complexity and level of detail
- Data type: Continuous measurements

4.2 Rationale for Feature Group Separation

Why not combine all 70 features immediately?

Mixing heterogeneous feature types creates theoretical and practical problems:

1. Distance metric incompatibility:

- Histogram data (HSV, LBP) represents probability distributions → appropriate distance metrics include Chi-square or Hellinger distance
- Continuous measurements (RGB moments, edge features) → Euclidean or Manhattan distance appropriate
- Applying a single distance metric (e.g., Euclidean) to all features assumes they are directly comparable, which is not theoretically justified

2. Feature dominance:

- Texture features (32 dimensions) would dominate simply due to dimensionality
- Edge features (2 dimensions) would have negligible influence
- Standardization addresses scale but not the fundamental issue of feature type differences

3. Interpretability:

- Clustering all features together produces results that are difficult to interpret
- Separating by feature type allows us to understand: “Do albums cluster by color? By texture? By complexity?”

Our solution: Analyze each feature group separately using appropriate distance metrics, then combine insights for supervised learning.

5 Unsupervised Learning: Feature Group Clustering

The core of our preliminary analysis involves clustering within each feature group to:

1. Identify natural visual style groupings using theoretically appropriate distance metrics
2. Assess the discriminative power of each feature group
3. Create interpretable cluster labels that can be used as features in supervised learning
4. Reduce dimensionality while preserving meaningful variation

This approach follows a variance decomposition framework:

- **Within-group variance:** How similar are albums based on one visual aspect (edges, texture, color)?
- **Between-group variance:** How do these aspects combine to create overall visual styles?

5.1 General Clustering Methodology

For each feature group, we follow a systematic process:

Step 1: Distribution Analysis

Examine raw feature distributions: skewness, kurtosis, outliers, normality tests (Shapiro-Wilk)

Step 2: Preprocessing Decision

Test multiple preprocessing methods:

- StandardScaler (z-score normalization)
- RobustScaler (median and IQR-based)
- PowerTransformer (Yeo-Johnson)
- QuantileTransformer

Select method that best reduces skewness and creates symmetric distributions.

Step 3: Distance Metric Selection

Test multiple distance metrics appropriate for the data type:

- Validate using cophenetic correlation (structure preservation in hierarchical clustering)
- Compare using silhouette scores and Davies-Bouldin index
- Select metric with strongest theoretical justification and empirical performance

Step 4: Optimal K Determination

Use internal validation indices:

- Silhouette score (higher is better)
- Davies-Bouldin index (lower is better)
- Visual inspection of elbow plots
- Balance statistical fit with interpretability

Step 5: Cluster Interpretation

For each cluster:

- Compute cluster means in feature space
- Identify distinguishing features (largest deviations from overall mean)
- Inspect representative album covers near cluster centroids
- Assign qualitative labels (e.g., “minimalist,” “complex,” “warm colors”)

5.2 Edge Features Clustering

5.2.1 Data Properties

- **Features:** `sobel_mean`, `edge_density`
- **Data type:** Continuous measurements (ratio scale)
- **Correlation:** $r = [\text{value from analysis}]$
- **Interpretation:**
 - Higher values $\rightarrow$ more edges, more visual complexity
 - Lower values $\rightarrow$ fewer edges, simpler/minimalist designs

5.2.2 Preprocessing

Method selected: PowerTransformer (or StandardScaler, based on empirical results)

Justification:

- Distribution analysis showed [describe skewness]
- PowerTransformer reduced skewness from [before] to [after]
- Created more symmetric distributions suitable for distance calculations

5.2.3 Distance Metric

Candidates tested: Euclidean, Manhattan (cityblock), Cosine, Correlation

Selected: Euclidean distance

Justification:

- **Absolute values matter:** An album with strong edges (high sobel mean) is fundamentally different from one with weak edges, not just a scaled version
- **Linear relationship:** More edges = more different (proportional dissimilarity)
- **Empirical validation:** Highest cophenetic correlation ($c = [\text{value}]$)
- **Best clustering quality:** Highest average silhouette score across $k \in \{2, 3, 4\}$

Alternatives rejected:

- Cosine distance: Ignores magnitude, focuses only on pattern (ratio) $\rightarrow$ inappropriate when absolute edge strength matters
- Correlation distance: Similar issue to cosine

5.2.4 Optimal K

Result: $K = 2$

Evidence:

- Silhouette score = 0.550 (> 0.5 indicates good separation)
- Davies-Bouldin index = 0.637 (lowest at $K = 2$, increases monotonically for $K > 2$)
- Clear visual separation in 2D scatter plot
- Natural interpretation as a dichotomy

5.2.5 Cluster Interpretation

Cluster 0: Minimalist Covers

- Low sobel mean, low edge density
- Visual characteristics: Solid colors, simple typography, clean portraits, uncluttered layouts
- Style: Professional, modern, minimalist aesthetic

Cluster 1: Complex Covers

- High sobel mean, high edge density
- Visual characteristics: Detailed artwork, collages, busy compositions, intricate patterns
- Style: Artistic, visually dense, high information content

Key finding: Edge features create well-separated, interpretable clusters with strong silhouette scores, indicating **high discriminative power**.

5.3 Texture Features Clustering

5.3.1 Data Properties

- **Features:** 32 LBP histogram bins
- **Data type:** Frequency counts (sum ≈ 0.125 per image, not normalized to 1)
- **Sparsity:** 0% (all bins contain non-zero values)
- **Distribution:** Highly skewed, dominated by bins 0 and 31

5.3.2 Preprocessing

Method selected: No preprocessing (raw frequencies)

Justification:

- LBP features are not compositional (do not sum to 1)
- Raw frequencies preserve magnitude information
- Chi-square distance naturally handles different bin scales through normalization: $d = \sum \frac{(p_i - q_i)^2}{p_i + q_i}$

5.3.3 Distance Metric

Candidates tested: Euclidean (standardized), Cosine, Chi-square

Selected: Chi-square distance

Justification:

- **Data type:** LBP features are histogram data (comparing distributions of texture patterns)
- **Standard in computer vision:** Chi-square is the established metric for comparing histograms
- **Theoretical basis:** Chi-square distance normalizes by bin frequencies, emphasizing differences in rare patterns
- **Empirical validation:** Highest cophenetic correlation ($c = 0.873$) compared to:
 - Euclidean (standardized): $c = 0.858$
 - Cosine: $c = 0.815$

5.3.4 Optimal K

Result: $K = 3$

Evidence:

- Davies-Bouldin index = 0.332 (lowest at $K = 3$)
- Cannot compute silhouette score with custom distance metric
- Used Davies-Bouldin as primary criterion

5.3.5 Cluster Interpretation

Severe cluster imbalance detected:

- Cluster 0: 296 albums (94%)
- Cluster 1: 18 albums (6%)
- Cluster 2: 1 album (0.3%)

Interpretation: The vast majority of album covers share similar texture patterns, dominated by specific LBP bins (bins 0 and 31). Texture features do **not** create meaningful subgroups in our dataset—most covers are texturally homogeneous with only rare outliers.

Key finding: Texture features have **low discriminative power** for distinguishing album covers. This suggests texture may not be a useful predictor in supervised learning.

5.4 RGB Features Clustering

[To be completed by teammate. Expected to follow similar methodology: distribution analysis → preprocessing selection → distance metric testing → optimal K determination → cluster interpretation.]

5.5 HSV Features Clustering

[To be completed by teammate. Given that HSV features are histograms, we expect Chi-square or similar histogram-specific distance metrics to be appropriate.]

5.6 Comparative Summary

Table 1 summarizes the clustering results across feature groups.

Table 1: Clustering results by feature group				
Feature Group	Optimal K	Distance Metric	Quality	Power
Edge (2 features)	2	Euclidean	Sil = 0.550	HIGH
Texture (32 features)	3	Chi-square	Imbalanced	LOW
RGB (12 features)	TBD	TBD	TBD	TBD
HSV (24 features)	TBD	TBD	TBD	TBD

Key insights so far:

- Edge features are highly discriminative (clear two-cluster structure)
- Texture features are not discriminative (one dominant cluster)
- Feature groups have different predictive potential

6 Transition to Supervised Learning

Once all feature groups are clustered, we will transition to supervised learning to predict Billboard 200 appearance.

6.1 Creating Meta-Features

After clustering all four feature groups, each album will have:

- Original 70 features (continuous and histogram values)
- 4 cluster assignment variables: `edge_cluster`, `texture_cluster`, `rgb_cluster`, `hsv_cluster`

These cluster labels represent **interpretable visual style categories** and serve as “informed labels” derived from unsupervised analysis.

6.2 Feature Selection Strategy

Based on clustering results, we will select features using:

1. Discriminative power from clustering:

- Prioritize features from groups with strong cluster separation (e.g., edge features)
- Consider dropping or dimension-reducing features from weak groups (e.g., texture features)

2. Variance contribution:

- Within each cluster, identify features with high between-cluster variance
- Drop near-constant or redundant features

3. Correlation analysis:

- Remove highly correlated features within each group
- Keep one representative from correlated feature sets

Candidate feature sets for supervised models:

- **Option 1:** Cluster labels only (4 categorical features, one-hot encoded)
- **Option 2:** Selected raw features (top 15–20 most discriminative) + cluster labels
- **Option 3:** All 70 features + cluster labels (with regularization)

6.3 Candidate Models

We will test multiple model classes:

6.3.1 Baseline: Logistic Regression

- L1 (Lasso) or L2 (Ridge) regularization
- Interpretable coefficients
- Establish performance floor

6.3.2 Tree-Based Models

- Random Forest
- Gradient Boosting (XGBoost or LightGBM)
- Advantages: Handle non-linear relationships, provide feature importance

6.3.3 Neural Network

- Architecture: Shallow network (2–3 hidden layers)
- Input: Selected features or full 70-feature vector
- Output: Binary classification (hit vs. non-hit)
- Handle class imbalance with weighted loss function
- Expected to be final model given flexibility and performance

6.4 Data Splitting Strategy

To preserve prediction integrity and ensure that the model does not overfit, this study has a designated validation set (20%) that will not be tested on until the final model has been developed. This ensures that the data is not overused and, as a result, does not overfit. The remaining data will be the “sandbox” dataset, where another 80/20 split will be applied. After each model test, the data is free to be scrambled and resplit before testing a new model. This ensures that multiple models can be tested while a holdout set remains untouched.

Specific splitting procedure:

1. **Protected validation set (20%):**

- Set aside *first*, never touched during model development
- Used only for final evaluation of selected model
- Ensures unbiased performance estimate

2. Sandbox dataset (80%):

- Used for all model development and tuning
- Can be re-split multiple times for different experiments
- Typical sandbox split: 80% train, 20% validation

Temporal considerations:

- Ideally, train on older albums (e.g., 2018–2022) and test on newer albums (e.g., 2023–2024)
- Mimics real prediction scenario: predict success of newly released albums
- Addresses potential non-stationarity in visual trends over time

6.5 Handling Class Imbalance

Billboard 200 appearances are relatively rare compared to all album releases. We will address class imbalance through:

- **Class-weighted loss functions:** Assign higher penalty to misclassifying hits
- **Sampling techniques:** Oversample minority class (hits) or undersample majority class (non-hits)
- **Evaluation focus:** Emphasize precision-recall metrics rather than accuracy

6.6 Evaluation Metrics

Primary metrics:

- **ROC-AUC:** Overall discrimination ability across all thresholds
- **Precision-Recall AUC:** More appropriate for imbalanced data than ROC-AUC

Secondary metrics:

- **Precision @ Top K:** If we predict the top 100 albums most likely to chart, how many actually do?
- **Calibration:** Are predicted probabilities well-calibrated? (Calibration plots)
- **Confusion matrix:** Understand false positive vs. false negative trade-offs

Interpretability:

- Logistic regression: Examine coefficients
- Tree models: Feature importance scores
- Neural networks: Partial dependence plots, SHAP values (if time permits)

7 Assumptions and Limitations

7.1 Assumptions

1. **Cover art contains signal:** We assume that visual features of album covers contain some predictive information about chart success, even if not causal.
2. **Feature extraction captures relevant information:** Our hand-crafted features (color, texture, edges) capture the visual characteristics that matter for success. Deep learning embeddings might capture additional information but are beyond our current scope.
3. **Temporal stability:** While visual trends change over time, we assume that within our analysis window, the relationship between visual features and success is reasonably stable.

7.2 Limitations

1. **Unobserved confounders:**
 - Artist popularity and existing fanbase
 - Marketing budget and promotional campaigns
 - Playlist placements and algorithmic promotion
 - Social media buzz and viral effects
 - Label backing and industry connections

These factors are likely more important than cover art but are unobserved in our data. Our models can only capture the *marginal* contribution of visual design, not its causal effect.

2. **Selection bias:**
 - Professional cover art is already optimized by designers
 - We do not observe “bad” cover art that was rejected before release
 - Our models learn patterns among covers that made it to market
3. **Feature engineering limitations:**
 - Hand-crafted features may miss important aspects: typography quality, facial expressions, brand recognition, cultural symbols

- Deep CNN embeddings would capture richer representations but are harder to interpret

4. **Sample size for prototyping:**

- Current analysis uses 315 albums for development
- Full dataset will be larger but still limited compared to deep learning standards

5. **Temporal non-stationarity:**

- Visual trends evolve (e.g., minimalist trend in 2020s)
- A style associated with hits in 2018 may not work in 2024
- Models may degrade over time

8 Anticipated Results and Applications

8.1 Expected Findings

Based on preliminary unsupervised analysis, we expect:

1. **Modest but non-trivial predictive power:**

- Visual features alone will not perfectly predict hits (too many confounders)
- But we expect ROC-AUC significantly better than random (>0.6–0.65)
- Suggests cover art contains *some* signal about commercial potential

2. **Edge features more predictive than texture:**

- Edge features showed strong clustering (high discriminative power)
- Texture features showed weak clustering (low discriminative power)
- Expect edge complexity to be more important in supervised models

3. **Certain visual styles over-represented among hits:**

- Preliminary hypothesis: Minimalist, high-contrast covers may correlate with mainstream pop success
- Complex, artistic covers may correlate with niche genres less likely to chart widely
- Will test these hypotheses in supervised phase

4. **Dimensionality reduction feasible:**

- Within each feature group, a small number of features or principal components will capture most useful variation
- Many raw features will be redundant or uninformative

8.2 Applications

If our models demonstrate predictive power, results could inform:

- **A&R and marketing strategy:**
 - Designers can test multiple cover concepts and estimate chart potential
 - Labels can make more informed decisions about visual branding
- **Understanding visual trends in music:**
 - Quantify how visual styles correlate with commercial success
 - Track evolution of effective visual design over time
- **Broader research on visual marketing:**
 - Methodology could extend to book covers, movie posters, product packaging
 - Demonstrates value of quantitative analysis of visual design

8.3 Interpreting Results Carefully

We emphasize:

- Our findings will be **correlational, not causal**
- Cover art does not *cause* chart success; it is one component of a complex system
- Successful albums likely have both good cover art *and* strong music, marketing, artist appeal, etc.
- Our goal is to quantify whether visual design contains **predictive information**, not to claim it is a primary driver of success

9 Summary and Next Steps

9.1 Current Status

- ✓ Edge features: Clustered ($K = 2$, high discriminative power)
- ✓ Texture features: Clustered ($K = 3$, low discriminative power)
- RGB features: In progress (teammate)
- HSV features: In progress (teammate)
- Merge with Billboard 200 data
- Supervised learning phase

9.2 Immediate Next Steps

1. Complete RGB and HSV feature clustering
2. Merge visual features with Billboard 200 chart data to create binary labels
3. Conduct feature selection based on clustering results
4. Build baseline logistic regression model
5. Develop and tune tree-based and neural network models
6. Evaluate final model on protected validation set
7. Interpret results and write final report